



Chapter 3

DFA Revision

Automata

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Exercise

Determine whether the following automaton is a DFA or an NFA:

- $q_0 \xrightarrow{a} q_1$
- $q_0 \xrightarrow{a} q_2$



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Suggested Answer It is an NFA because one state has multiple transitions for the same input symbol.





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An automaton contains:

- $q_0 \xrightarrow{\varepsilon} q_1$
- $q_1 \xrightarrow{a} q_2$

Identify all ε -transitions.



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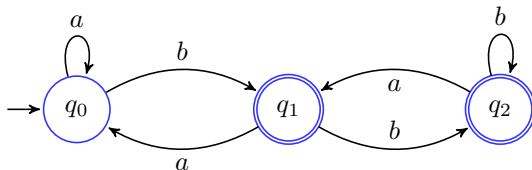
Identify all ε -transitions.

Suggested Answer $q_0 \xrightarrow{\varepsilon} q_1$

Exercise

Let $\Sigma = \{a, b\}$

- 1 Does the given DFA accept the string "ab"?
- 2 Does the DFA accept "aba"?
- 3 Does the DFA accept "bbb"?
- 4 Does the DFA accept the empty string ε ?
- 5 Does the DFA accept "aab"?
- 6 Does the DFA accept "baba"?
- 7 Does the DFA accept "aaab"?
- 8 Does the DFA accept "abb"?
- 9 Does the DFA accept "ba"?
- 10 Does the DFA accept "abab"?



Exercise



Convert the following NFA to DFA:

- $q_0 \xrightarrow{a} q_0$
- $q_0 \xrightarrow{a} q_1$
- $q_1 \xrightarrow{b} q_2$

Final state: q_2

Exercise



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Let $\Sigma = \{a, b, c\}$

- Construct an DFA for the regex $E_1 = (a + b)^*ab$,
- Construct an DFA for $E_2 = a(b + c)^*$,
- Construct an DFA for $E_3 = (ab + ba)^*$,
- Construct an DFA for $E_4 = (a + bc)^*$,
- Construct an DFA for $E_5 = a^*b^*$,
- Construct an DFA for $E_6 = (a + b)^*aa$,
- Construct an DFA for $E_7 = ab(a + b)^*$,
- Construct an DFA for $E_8 = (ab)^* + (ba)^*$,
- Construct an DFA for $E_9 = (a + b)^*(ab)(a + b)^*$,
- Construct an DFA for $E_{10} = (a + b)^*b(a + b)^*b$.



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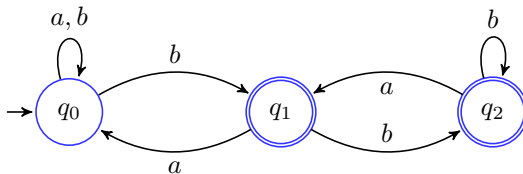
Complex NFA to DFA

Let $\Sigma = \{a, b, c\}$

- ① Construct an DFA for strings containing at least one 'a'
- ② Construct an DFA for strings ending with 'b'
- ③ Construct an DFA for strings starting with 'a'
- ④ Construct an DFA for strings containing substring "ab"
- ⑤ Construct an DFA for strings not containing "bb"
- ⑥ Construct an DFA for strings with exactly two 'a' characters
- ⑦ Construct an DFA for strings with even number of 'b'
- ⑧ Construct an DFA for strings with length ≥ 2
- ⑨ Construct an DFA for strings containing "aba"
- ⑩ Construct an DFA for strings without 'a'.

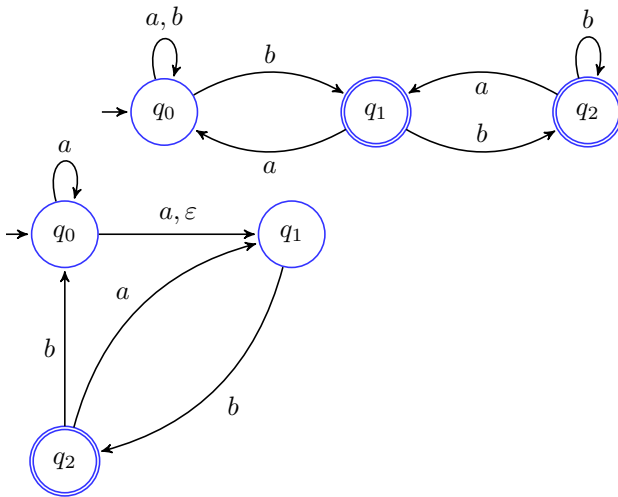
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Let $\Sigma = \{a, b\}$, convert NFA to DFA



Exercise

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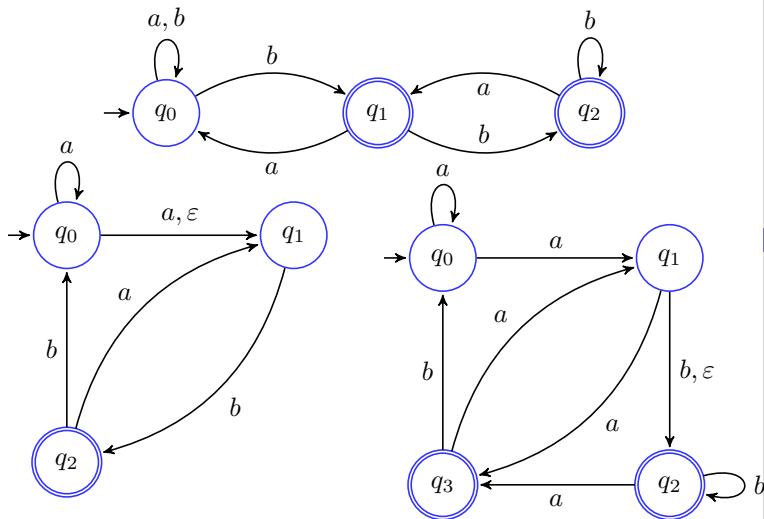
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Which strings are (not) accepted by the automata?

① abab

② ...

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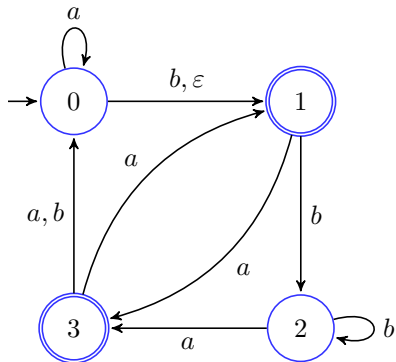
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